



Uber Database System



<i>Trainee Name</i>	<i>Instructor Name</i>	<i>Assistant Name</i>
<i>Maysun Salah</i>	<i>Majd Nasser</i>	<i>Abdulmajed Almutairi</i>

Introduction

Uber is a global ride-sharing platform that connects passengers with nearby drivers through a mobile application. The system manages real-time trip requests, dynamic pricing, driver availability, and secure payment processing. This database design models four core entities of the Uber platform: Users, Drivers, Trips, and Payments, linked together through foreign key relationships.

Tables Overview

Table	Main Fields	Description
Users	id, full_name, email, gender, country_code	Stores registered passenger accounts
drivers	id, full_name, license, rating, is_available	Stores driver profiles and availability
trips	id, user_id, driver_id, distance, status, created_at	Records each ride and links user to driver
payments	id, trip_id, amount, method, status	One payment record per completed trip

Database Schema Diagram

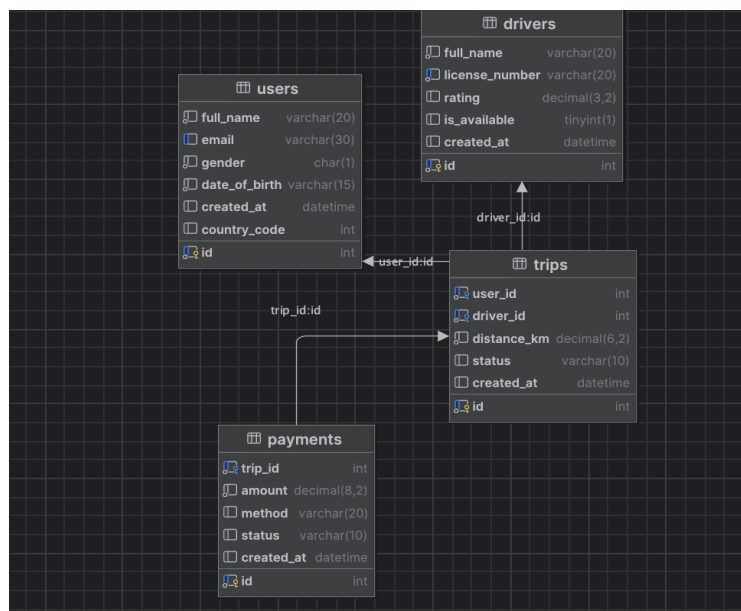


Figure 1 — Entity Relationship Diagram for the Uber Database

Queries

```
-- create the database

create database uber;

-- select the database to use it

use uber;

-- users table (stores passenger info)

create table users (
  id          int primary key,
  full_name   varchar(20) not null,
  email       varchar(30) unique,
  gender      char(1) not null check(gender = 'm' or gender = 'f'),
  date_of_birth varchar(15) not null,
  created_at  datetime,
  country_code int
);

-- drivers table (stores driver profiles)

create table drivers (
  id int primary key,
  full_name varchar(20) not null,
  license_number varchar(20) unique not null,
  rating decimal(3,2) default 5.00,
  is_available boolean default true,
  created_at datetime
);

-- trips table (each ride links a user and a driver)

create table trips (
  id int primary key,
  user_id int not null,
  driver_id int not null,
  distance_km decimal(6,2) not null,
  status varchar(10) check(status = 'pending' or status = 'completed' or status = 'cancelled'),
  created_at datetime,
```

```

-- link trip to user and driver

foreign key (user_id) references users(id),
foreign key (driver_id) references drivers(id)
);

-- payments table (one payment per trip)

create table payments (
  id int primary key,
  trip_id int not null unique,
  amount decimal(8,2) not null,
  method varchar(20) check(method = 'cash' or method = 'card' or method =
'applePay'),
  status varchar(10) check(status = 'pending' or status = 'paid' or status =
'refunded'),
  created_at datetime,

  -- link payment to trip

foreign key (trip_id) references trips(id));

-- add default timestamps after table creation

alter table users modify created_at datetime default current_timestamp;
alter table drivers modify created_at datetime default current_timestamp;
alter table trips modify created_at datetime default current_timestamp;
alter table payments modify created_at datetime default current_timestamp;

-- insert users data

insert into users (id, full_name, email, gender, date_of_birth) values
(1, 'maysun alharbi', 'maysun@gmail.com', 'f', '2001-07-09'),
(2, 'muath salah', 'muath@gmail.com', 'm', '2002-05-05'),
(3, 'luluah', 'lulu@gmail.com', 'f', '2006-08-20'),
(4, 'saleh', 'saleh@gmail.com', 'm', '2003-01-15'),
(5, 'laten', 'laten@gmail.com', 'f', '2002-09-30');

-- insert drivers data

insert into drivers (id, full_name, license_number, rating, is_available) values
(1, 'faisal ', 'SA-001', 4.90, true),
(2, 'waeel', 'SA-002', 4.75, true),
(3, 'yousef', 'SA-003', 4.60, false),

```

```
(4, 'omar', 'SA-004', 4.85, true),  
(5, 'adel', 'SA-005', 4.50, false);
```

-- insert trips data

```
insert into trips (id, user_id, driver_id, distance_km, status) values  
(1, 1, 1, 12.5, 'completed'),  
(2, 2, 2, 7.3, 'completed'),  
(3, 3, 1, 20.0, 'completed'),  
(4, 4, 3, 5.0, 'cancelled'),  
(5, 5, 4, 15.8, 'completed');
```

-- insert payments data

```
insert into payments (id, trip_id, amount, method, status) values  
(1, 1, 45.00, 'card', 'paid'),  
(2, 2, 27.50, 'applePay', 'paid'),  
(3, 3, 72.00, 'cash', 'paid'),  
(4, 4, 0.00, 'card', 'refunded'),  
(5, 5, 56.80, 'applePay', 'paid');
```

-- update users

```
update users set email = 'mays@gmail.com' where id = 1;  
update users set full_name = 'moaad salah' where id = 2;  
select *from users;
```

-- update drivers

```
update drivers set rating = 4.95 where id = 1;  
select *from drivers;
```

-- update payments

```
update payments set status = 'paid' where id = 4;  
select *from payments;
```

-- delete some data

```
delete from payments where trip_id = 4;
```

```
--update users on country_code filed
```

```
update users set country_code = 966 where id = 1;
```

```
update users set country_code = 966 where id = 2;
```

```
update users set country_code = 977 where id = 3;
```

```
update users set country_code = 959 where id = 4;
```

```
update users set country_code = 974 where id = 5;
```

```
select *from users;
```